Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 32)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Describe how sulfur can be removed from coal **before** combustion.

[2 marks]

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0 1 . 2 Describe the process of dry flue gas desulfurisation and explain how the product can be used.

[3 marks]

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0 1 . 3 In wet flue gas desulfurisation the flue gases are bubbled through sodium sulfite solution. Explain **one** advantage of this method over dry FGD.

[2 marks]

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0 1 . 4 Explain how each of the following reduces NOx emissions.

[3 marks]

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Fluidised bed combustion .....

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Catalytic converters .....

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Urea sprays .....

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0 2 Table 1 shows some properties of three pollutants.

Table 1

Pollutant	State at 20 °C	Solubility in water	Solubility in lipids	Environmental half-life
DDT (insecticide)	Solid	Very low	High	About 15 years in soil
Nitrate ions	Dissolved ion	Very high	Very low	Not applicable (not degraded)
Hydrogen cyanide	Gas (denser than air)	High	Low	Hours

0 2 . 1 Define the term **environmental half-life**.

[1 mark]

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0 2 . 2 Use Table 1 to explain why DDT is more likely than nitrate ions to biomagnify in a food chain.

[3 marks]

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0 2 . 3 Suggest why the release of hydrogen cyanide at Bhopal, India, in 1984 caused deaths very close to the source rather than over a wide area.

[2 marks]

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0 2 . 4 Explain how the high water solubility of nitrate ions affects the area over which they cause pollution and the concentration of the pollution.

[2 marks]

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0 2 . 5 Give **two** examples of pollutants that are forms of energy rather than substances.

[2 marks]

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0 3 Table 1 shows some techniques used to locate mineral deposits.

0 3 . 1 Complete Table 1 by adding the name of the technique or a description of how it works.

[4 marks]

Table 1

Technique	How it works
Gravimetry	
	Detects rocks that are more magnetic, such as magnetite iron ore or ores of tungsten and cobalt
Seismic survey	
	Measures how difficult it is for electricity to pass through rocks; sedimentary rocks with high water content have lower values than igneous rocks

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0 3 . 2 Explain why trial drilling is usually carried out only after remote sensing surveys.

[3 marks]

0 3 . 3 Explain how infrared spectroscopy from satellites can help identify mineral deposits.

[2 marks]

0 3 . 4 Give **one** advantage of portable field analysis equipment such as X-ray fluorescence.

[1 mark]

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0 4 A coal-mine spoil heap on a hillside has been abandoned. The spoil contains shale with some iron pyrites and residual coal.

0 4 . 1 Explain **three** problems this spoil heap could cause and describe how each could be managed.

[6 marks]

0 4 . 2 Explain why deep layers of soil are applied when a spoil heap containing toxic materials is restored.

[2 marks]

0 4 . 3 Explain why an old steel-works spoil heap may be valuable for wildlife conservation.

[2 marks]

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0 5 A student used 0.5 m x 0.5 m grid quadrats to survey ground flora in two meadows, one grazed and one ungrazed. Table 1 shows the results for the grazed meadow (10 quadrats).

Table 1

Species	Number of individuals
Ribwort plantain	48
Perennial ryegrass	120
White clover	36
Daisy	24
Creeping buttercup	12

0 5 . 1 Explain why a 0.5 m x 0.5 m quadrat was chosen rather than a 10 cm x 10 cm or 100 m x 100 m quadrat.

[2 marks]

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0 5 . 2 Calculate Simpson's Index of Biodiversity (D) for the grazed meadow using D = N(N-1) / Σn(n-1). Show your working.

[3 marks]

Answer

0 5 . 3 The ungrazed meadow had D = 1.8 with the same number of species. Explain what this shows.

[2 marks]

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0 5 . 4 Explain why a point quadrat would be better than a frame quadrat in tall vegetation.

[2 marks]

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0 5 . 5 Give **one** limitation of using the DAFOR abundance scale.

[1 mark]

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0 6 A rural community in Pakistan is considering a low-head helical (Archimedes screw) turbine on a river rather than joining a high-head dam scheme.

0 6 . 1 Explain why low-head turbines are more suitable than high-head HEP schemes for many rural areas.

[5 marks]

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0 6 . 2 Suggest **one** reason why such schemes may not be installed even at suitable sites.

[1 mark]

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0 6 . 3 The maximum flow rate at the site is 6.5 m^3 s^{-1} giving an output of 100 kW at 90% turbine efficiency. At 10% of maximum flow the efficiency falls to 40%. Calculate the output at 10% of maximum flow.

[2 marks]

Answer ☐ kW

0 6 . 4 Outline **two** ways a high-head reservoir can affect the river downstream.

[2 marks]

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0 7 . 1 Which chemical form of mercury is most easily absorbed and biomagnified in food chains? Tick **one** box.

[1 mark]

A Elemental (liquid) mercury ☐

B Inorganic mercury compounds ☐

C Organic (methyl) mercury ☐

D Mercury vapour ☐

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0 7 . 2 Explain why anaerobic sediments increased the hazard of the inorganic mercury discharged at Minamata.

[2 marks]

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0 7 . 3 Explain why heavy metal wastes are stored under alkaline conditions.

[2 marks]

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0 8 Table 1 shows the textural composition of four soil samples.

Table 1

Soil sample	Sand / %	Silt / %	Clay / %
A	85	10	5
B	40	40	20
C	20	25	55
D	10	80	10

0 8 . 1 Identify the sample that is a loam and explain why loams are ideal for cultivating most crops.

[3 marks]

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0 8 . 2 Explain why sample A is likely to have the highest permeability but the lowest nutrient retention.

[3 marks]

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0 8 . 3 Compare the thermal capacity and root penetration of soils A and C.

[2 marks]

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0 8 . 4 Describe how the texture of a soil sample could be determined by sedimentation.

[2 marks]

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0 9 A dam and reservoir are to be built on a river to supply water and regulate flow.

0 9 . 1 Explain **three** effects the reservoir may have on wildlife in and around the river.

[6 marks]

0 9 . 2 Explain why decomposition of flooded vegetation in a new reservoir contributes to climate change.

[2 marks]

0 9 . 3 Explain how a river-regulation reservoir reduces both flooding and water shortages downstream.

[2 marks]

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1 0 . 1 Explain how organic wastes and inorganic nutrients cause water pollution and evaluate the effectiveness of the methods used to control them.

[9 marks]

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Extra space

1 0 . 2 Give **one** reason why algae cause eutrophication problems more quickly than rooted plants when nutrients are added to a lake.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably.

[25 marks]

OR

1 1 . 2 Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS